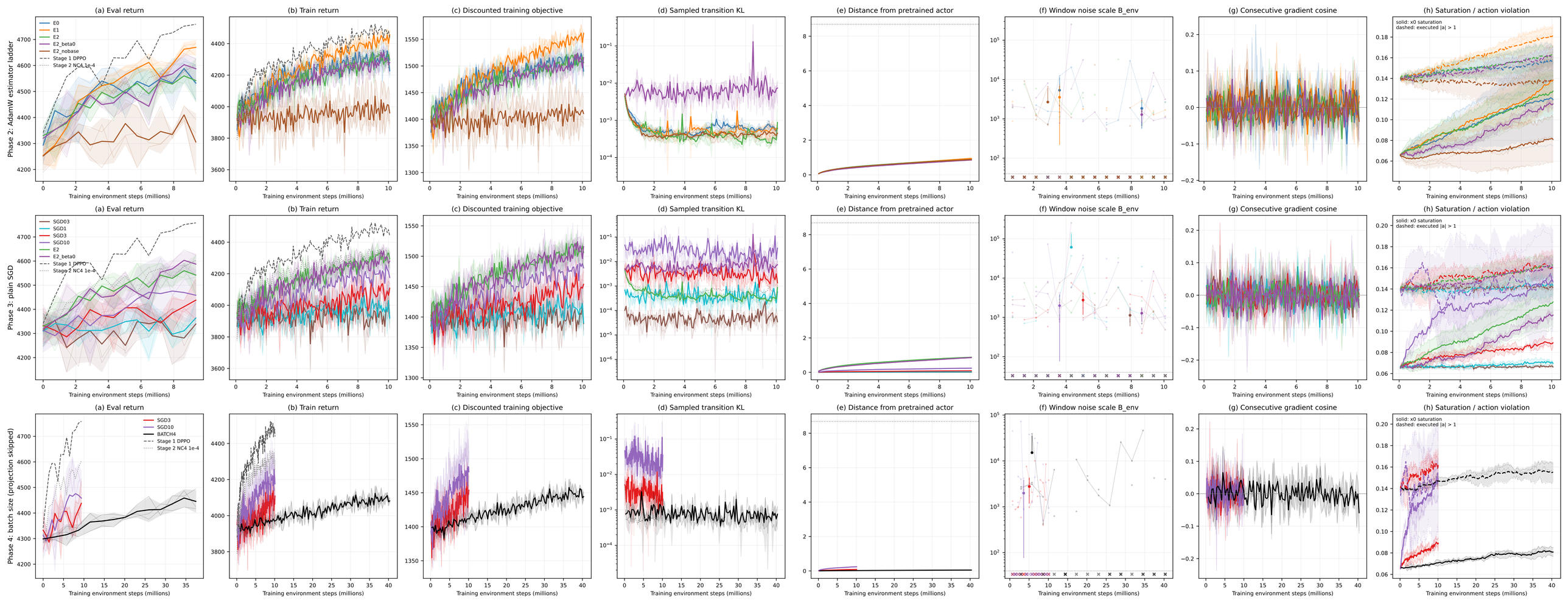

Phase 2: AdamW estimator ladder
Phase 3: plain SGD
Phase 4: batch size (projection skipped)
(a) Eval return
(b) Train return
(c) Discounted training objective
(d) Sampled transition KL
(e) Distance from pretrained actor
(f) Window noise scale B_env
(g) Consecutive gradient cosine
(h) Saturation / action violation
E0
E1
E2
E2_beta0
E2_nobase
Stage 1 DPPO
Stage 2 NC4 1e-4
SGD03
SGD1
SGD3
SGD10
BATCH4
solid: x0 saturation
dashed: executed |a| > 1
Training environment steps (millions)